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1) $N=32$ $N \times N$ FP32 matrix operations.

For every $C[i][j] = \sum_{k=0}^{N} A[i][k] \cdot B[k][j]$, Row i of A and column j of B accessed only once. Other elements of B are not accessed, so exactly $N=32$ elements of B are accessed once.

Total element accesses: For one element $C[i][j]$, $2 \cdot 32$ elements accessed. 32×32 elements $\Rightarrow$ $2 \cdot 32^3 = 2N^3$ accesses made to A and B.

DRAM traffic: FP32 $\Rightarrow$ $4 \cdot 2 \cdot 32^3$ bytes for A and B, $4 \cdot 32^2$ bytes for storing to C $\Rightarrow$ $2 \cdot 32^3 \cdot 4 + 32^2 \cdot 4$ total DRAM traffic. $= 4(2N^3 + N^2)$ bytes

2) Tiled GEMM $T=8$. 8×8 tiles. 32^2 elements total, 8^2 elements per tile.
2^6 2^{10} 2^6

$\Rightarrow 2^4 = 16$ tiles. Tiles used only once

DRAM loads for A and B: $2^6 \cdot 2^4 = 32^2$ loads per matrix

2 matrices $\Rightarrow$ $2 \cdot 32^2$ DRAM loads for A and B.

Total DRAM traffic: $\underset{(A \text{ and } B)}{2 \cdot 32^2} + \underset{(C)}{32^2} = 3 \cdot 32^2$ FP32 $\Rightarrow$ $4 \cdot 3 \cdot 32^2$ bytes $= 12N^2$ bytes

3) Ratio of naive traffic to tiled traffic: $\frac{4(2N^3+N^2)}{4 \cdot 3N^2}$ $\frac{2N^3+N^2}{3N^2} = \frac{2}{3}N + \frac{1}{3}$ $= \underline{\frac{2N+1}{3}}$

Explanation: Assuming unbounded shared memory, matrices A and B only need to be loaded once in tiled blocks, and C is still accessed only once as well, for $3 \cdot N^2$ total accesses. Tiling reduced the number of accesses by a factor of N, i.e. tiled traffic is $O(N^2)$. Ratio of naive traffic to tiled is $O(N^3)/O(N^2) = O(N)$.

4) Execution time: 10 TFOP/s compute, 320 GB/s BW.

Naive: $4 \cdot 2 \cdot 32^3 + 4 \cdot 32^2$ bytes, ~~302~~ 320 GB/s

$$\Rightarrow \frac{4 \cdot 2 \cdot 32^3 + 4 \cdot 32^2}{320 \cdot 2^{30}} = 7.749 \cdot 10^{-7} \text{ seconds} = 0.7749 \text{ us}$$

or

$2 \cdot 32^3$ FLOPS, 10 TFOP/s $= 10 \cdot 2^{40}$ FLOP/s

$$\Rightarrow \frac{2 \cdot 32^3}{10 \cdot 2^{40}} = 5.960 \cdot 10^{-9} \text{ seconds} = 5.96 \cdot 10^{-3} \text{ us}$$

Total execution time: $7.749 \cdot 10^{-7} + 5.960 \cdot 10^{-9}$

$= 7.809 \cdot 10^{-7}$ s $= 7.809 \cdot 10^{-1}$ µs.

or 780.9 ns

Tiled: $\frac{3 \cdot 32^2 \cdot 4}{320 \cdot 2^{30}} = 3.576 \cdot 10^{-8}$ s

Total Execution time: $3.576 \cdot 10^{-8} + 5.96 \cdot 10^{-9}$

$= 4.172 \cdot 10^{-8}$ s $= 41.7$ ns

Roofline ridgepoint: (31.25 FLOP/B, 10 TFLOP)

Naive AI: $\frac{2 \cdot 32^3}{4 \cdot (2 \cdot 32^3 + 32^2)} = 0.246$ FLOP/B $\Rightarrow$ Memory bound

Tiled AI: $\frac{2 \cdot 32^3}{4 \cdot 3 \cdot 32^2} = 5.333$ FLOP/B $\Rightarrow$ Memory bound, but still much better than naive GEMM.